PEARSON EDEXCEL INTERNATIONAL GCSE (9–1) · CHEMISTRY 4CH1 · PAPER 1C

A* Examination Guide

Built from 17 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the papers themselves, not from the specification. Every model answer uses the wording the mark schemes reward. Every warning comes from a REJECT, IGNORE or "0 marks if…" note that examiners actually applied.

1	How Paper 1C is built: structure, topic frequency, recurring 6-mark questions
2	Twenty mark-scheme rules that separate A* from A
3	States of matter, diffusion and purity
4	Mixtures, separation and chromatography
5	Atomic structure, isotopes and the Periodic Table
6	Groups 1, 7 and 0
7	Gases in the atmosphere and the % oxygen experiment
8	Chemical formulae, moles and every calculation type
9	Bonding, structure and properties
10	Reactivity, redox and rusting
11	Acids, alkalis and salt preparation
12	Chemical tests
13	Energetics
14	Rates of reaction
15	Reversible reactions and water of crystallisation
16	Organic chemistry: crude oil, alkanes, alkenes, polymers
17	Practical and graph skills
18	Final A* checklist

Papers analysed: Jan 2019 (legacy 4CH0, 120 marks); Jun 2019; Jan 2020; Nov 2020; Jan 2021; Jun 2021; Nov 2021; Jan 2022; Jun 2022; Jan 2023; Jun 2023; Jan 2024; Jun 2024; Jan 2025; Jun 2025; Nov 2025; Jun 2026. Throughout, papers are cited by the series they were sat in, e.g. (Jun 24).

1 How Paper 1C is built

Format: 2 hours, 110 marks, 9–12 questions, roughly 1 mark per minute. Questions start easy (states of matter, gases, separation) and get harder towards the end. The final questions are usually multi-step calculations (moles, enthalpy, water of crystallisation) or a 6-mark practical description. Multiple-choice items (A–D) are worth 1 mark each. You also need to know why the wrong options are wrong, because the distractors repeat.

Topic frequency across the 17 papers

Topic	Papers (of 17)	What is asked most
Mole calculations (any type)	17	Empirical formula, reacting masses, % yield, excess reagent, Ar from isotopes
Organic: alkanes / alkenes / isomers	17	Isomer definition and drawing, bromine water test, substitution vs addition, combustion
Bonding and structure explanations	16	Melting/boiling point explanations, dot-and-cross, diamond/graphite/C_{60}
Rates of reaction	14	Collision theory, sketching a second curve, tangent gradient, graph shape
Chemical tests (ions, gases, water)	14	6-mark identification of an unknown salt
States of matter / changes of state	14	Naming changes, particle description, state symbols
Reactivity series, displacement, redox	13	Order from data, observations, oxidising/reducing agent
Energetics ($Q = mc\Delta T$, ΔH)	12	Calorimetry calculation with sign, sources of error, polystyrene cup
Rusting and % oxygen in air	11	Conditions, galvanising explanation, % O_2 calculation
Salt preparation / crystallisation	11	4–6 mark method for pure, dry crystals
Polymers and disposal	10	Repeat unit, landfill vs burning problems
Isotopes and relative atomic mass	9	Definition, Ar to 1 d.p. or 3 s.f.
Water of crystallisation (find x)	9	Masses → moles → ratio; heat to constant mass
Chromatography	8	Method, pencil line, interpreting spots, Rf
Diamond / graphite / fullerene	8	6-mark comparison; graphite conducts and is soft
Crude oil and cracking	7	Fractional distillation description, cracking conditions and reasons
Group 1 and Group 7	10	Li/Na with water observations; halogen displacement

The recurring 6-mark (and 5-mark) questions

- **Identify an unknown compound by chemical tests:** Jan 20, Jan 21, Jun 21, Jan 24, Jun 24, Nov 25. One cation test and one anion test, three marks each.
- **Make pure, dry crystals of a salt:** Nov 20, Nov 21, Jan 22, Jan 23, Jun 19.
- **Diamond vs graphite (structure, bonding, conductivity, hardness):** Jan 20, Jun 22, Jun 25.
- **Halogen displacement to show order of reactivity:** Jun 23.
- **Fractional distillation and cracking:** Nov 25 (labelled diagram), Jun 26.
- **Ethane vs ethene with bromine (5 marks):** Jun 22, Jun 25.
- **Melting point comparison, giant ionic vs simple molecular (5 marks):** Jun 24, Jan 25, Nov 25.

Learn the model answers for these word for word. They account for roughly 15–20 marks on every paper.

2 Twenty mark-scheme rules that separate A* from A

1. **Covalent bond:** attraction between a shared pair of electrons and the *nuclei*. Writing "nucleus" (singular) loses the mark.
2. **Simple molecular substances:** any mention of breaking *covalent bonds* when they melt or boil scores 0 for the explanation. Say "weak intermolecular forces".
3. **Giant structures:** any mention of *intermolecular forces* or *molecules* in diamond, graphite or ionic lattices scores 0.
4. "More energy" is ignored. Write "**a large amount of energy** is needed to break the strong bonds".
5. **Collisions:** write "more collisions *per unit time*" or "more frequent collisions". "Greater chance of collisions" is ignored.
6. For **concentration, surface area or pressure**, never mention particles gaining energy (max 1). Energy belongs only to **temperature**.
7. **Polystyrene cup:** it is an insulator, so it "*reduces* heat loss". "Prevents heat loss" is rejected.
8. **Never round a final answer to 1 significant figure** unless the question asks for it ("any number of s.f. except 1"). Give 3 s.f. by default.
9. **ΔH must carry a sign**: negative for exothermic, positive for endothermic. A missing sign costs 1 mark.
10. Read thermometers to **1 decimal place**, e.g. 16.0, not 16.
11. **Galvanising:** zinc "oxidises / corrodes / reacts instead of iron". "Zinc rusts" is rejected; only iron rusts.
12. Rust is **(hydrated) iron(III) oxide**. "Iron oxide" is ignored and "iron(II) oxide" is rejected.
13. **-ine vs -ide:** chlorine is the element and chloride the ion. "Bromine displaces iodide" loses a mark.
14. **Hydrocarbon:** a compound of hydrogen and carbon *only*. "Hydrogen and carbon molecules" is rejected.
15. **Isomers:** same *molecular* formula, different *structural/displayed* formula. "General formula" or "chemical formula" is rejected.
16. **Homologous series:** write "*trend* in physical properties". "Similar physical properties" is rejected.

17. **Bromine water:** orange → colourless ("decolourised"). "Clear" is ignored and "discoloured" is rejected.
18. **Gas tests act on the gas.** Bubble the gas into limewater, and hold damp red litmus in the gas. Adding limewater or litmus to the solution scores 0.
19. **Acidify before ion tests** with the right acid: HNO_3 before $AgNO_3$, and HCl or HNO_3 before $BaCl_2$. Adding HCl before silver nitrate, or H_2SO_4 before barium chloride, loses the test.
20. **Crystals:** heating the solution to dryness caps the method at about 1–2 marks. Heat only to the crystallisation point, then cool.

3 States of matter, diffusion and purity

Change	Name	Change	Name
solid → liquid	melting	liquid → solid	freezing
liquid → gas	evaporation / boiling	gas → liquid	condensation
solid → gas	sublimation (iodine, CO_2)	"heating" and "cooling" are distractors, not names of changes	

State	Arrangement	Movement
Solid	regular, closely packed	vibrate about fixed positions
Liquid	close together, irregular/random	move around each other
Gas	random, far apart / widely spaced	move quickly and randomly

Particle diagrams: draw liquid particles touching, filling the bottom of the box, irregular. Draw gas particles far apart with none touching. Particles in a solid have the least energy.

MODEL ANSWER · WHY HEATING MAKES A LIQUID EVAPORATE FASTER (2)
Particles gain kinetic energy and move faster, so more particles have enough energy to overcome the intermolecular forces and escape from the surface.

State symbols for a change of state: condensation is $H_2O(g) \rightarrow H_2O(l)$, and freezing is $H_2O(l) \rightarrow H_2O(s)$.

Diffusion

- A crystal placed in water undergoes **dissolving** and then **diffusion**. The state change is $X(s) \rightarrow X(aq)$.
- **NH_3 / HCl tube (Jan 22, Nov 25):** the white ring of NH_4Cl forms *closer to the HCl end*. HCl has the higher M_r (36.5 vs 17), so its molecules move more slowly. The ring takes minutes to form because gas particles move in random directions and collide with air particles.
- Warmer water gives a faster result because particles move faster and meet sooner.

Pure or mixture?

- A pure substance melts at a **sharp, fixed** temperature. An impure substance melts **over a range**, at a lower temperature.
- **Physical test for pure water:** boils at exactly 100 °C, or freezes/melts at 0 °C.

- **Chemical test for water** (shows water is present, not that it is pure): anhydrous copper(II) sulfate turns white → blue, or cobalt chloride turns blue → pink.

4 Mixtures, separation and chromatography

Compound: two or more different elements chemically bonded together. **Mixture:** different substances that are *not* chemically joined.

Separate…	Method
insoluble solid from a liquid	filtration (the solid is the residue)
soluble solid from a solution	crystallisation
pure water (solvent) from a solution	simple distillation
liquids with different boiling points (ethanol/water, crude oil)	fractional distillation
dyes in an ink or food colouring	paper chromatography

- **Rock salt (Jan 19):** crush; warm and stir in water so the salt dissolves faster; filter, leaving the insoluble impurities as the residue; crystallise the filtrate. Water is the *solvent*, and the filtrate is a *solution*.
- **Condenser:** cools the vapour so it condenses into liquid.
- To dissolve a solid faster: heat, stir, or crush it.

MODEL ANSWER · CHROMATOGRAPHY METHOD (5)

1. Draw a start line in **pencil** near the bottom of the paper.
2. Put a small spot of each ink on the line.
3. Pour solvent (e.g. water) into a beaker to a depth **below** the start line.
4. Stand the paper in the beaker so the spots are **above** the solvent level.
5. Leave until the solvent has risen nearly to the top, then remove the paper, mark the solvent front and let it dry.

MODEL ANSWERS · THE TWO CLASSIC MISTAKES (4)

Baseline drawn in ink: the ink dissolves in the solvent and moves up the paper, interfering with the results. Pencil is insoluble.
Solvent above the spots: the dyes dissolve into the solvent in the beaker instead of travelling up the paper.

Interpreting a chromatogram

- **Same dye** means spots at the same height, i.e. the same Rf value.
- **Most soluble dye** travels furthest. **Insoluble dye** stays on the start line (Rf = 0).
- One spot that has moved means one dye. A spot that has *not* moved could still contain several dyes, because they have not separated (Jun 24, Jun 25).
- **Rf = distance moved by spot ÷ distance moved by solvent**, both measured from the start line. Rf is always less than 1.
- Reverse use: distance = Rf × solvent distance, e.g. 0.72 × 120 mm = 86 mm.

5 Atomic structure, isotopes and the Periodic Table

Particle	Relative mass	Relative charge	Location
proton	1	+1	nucleus
neutron	1	0	nucleus
electron	1/1836 (≈ 0.0005)	−1	shells

- Atomic number = number of protons. Mass number = protons + neutrons. In an ion, electrons ≠ protons.
- **Negative ion** (e.g. 7 p, 7 n, 10 e): it has more electrons than protons, and electrons are negative while protons are positive.
- **Group** = number of outer-shell electrons. **Period** = number of occupied shells. Configurations up to Ca: 2.8.8.2.
- Within a period, atomic number increases by 1 for each step to the right. If Q has 31 protons, the element two places to the right has 33.
- Elements in the same group react similarly because they have the same number of outer-shell electrons.
- Metals form **basic** oxides (Li, Na, Mg). Non-metals form **acidic** oxides (SO_2, CO_2).

MODEL ANSWER · ISOTOPES (2)
Atoms of the same element (same number of protons) with different numbers of neutrons. They have the same chemical properties because they have the same electronic configuration.

Relative atomic mass from isotopes

A_r = Σ(mass number × % abundance) ÷ 100, then round to the d.p. or s.f. asked.
Br: (79 × 52.8 + 81 × 47.2) ÷ 100 = 79.944 → **79.9** (3 s.f.) · Mg: (24 × 79.0 + 25 × 10.0 + 26 × 11.0) ÷ 100 = **24.3** · B: (10 × 20.6 + 11 × 79.4) ÷ 100 = **10.8** · C: **12.01** (2 d.p.)
Unrounded answers such as 79.944, when 3 s.f. were asked for, score only 2 of 3.

Avogadro calculations (1 mol = 6.0 × 10^{23} particles)

- Electrons in 1 mol of Mg atoms = 12 × 6.0 × 10^{23} = **7.2 × 10^{24}**
- Atoms in 1 mol of C_{60} = 60 × 6.0 × 10^{23} = **3.6 × 10^{25}**
- Molecules in 100 g F_2: 100 ÷ 38 = 2.63 mol, × 6.0 × 10^{23} = **1.6 × 10^{24}**
- Mass of one Mg atom = 24.32 ÷ 6.022 × 10^{23} = **4.039 × 10^{-23} g**
- Water molecules from 0.02 mol $CuSO_4 \cdot 5H_2O$: 0.1 mol, which is **6 × 10^{22}**

6 Groups 1, 7 and 0

Group 1 (Li, Na, K)

- **Observations with water** (any 2–3): fizzing/effervescence; floats; moves on the surface; gets smaller and disappears. Sodium also melts into a ball. Potassium gives a lilac flame. ("Gas given off" and "heat produced" are ignored.)
- Products: metal hydroxide + hydrogen, e.g. 2Li + $2H_2O$ → 2LiOH + H_2.
- The solution is **alkaline** because OH^- ions are present. Universal indicator turns blue/purple (pH 10–14), phenolphthalein turns pink and methyl orange turns yellow.

- Reactivity increases down the group.
- **Equal masses (Jun 26):** 0.25 g K is fewer moles than 0.25 g Na (A_r 39 vs 23), so it gives **less** hydrogen.
- $4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$. The product is ionic and basic, and the reaction is exothermic. Activation energy is the *minimum* energy needed for a reaction to happen.

Group 7 (halogens)

Halogen	State (room temp.)	Colour	Colour in aqueous solution
fluorine	gas	pale yellow	–
chlorine	gas	pale green	pale green / colourless
bromine	liquid	red-brown	orange / yellow
iodine	solid	grey-black (purple vapour)	brown
astatine (predict)	solid	black	–

- Reactivity **decreases** down the group as the atoms get bigger. Boiling point **increases** down the group: molecules get larger, so intermolecular forces get stronger and more energy is needed to overcome them.
- Fluorine with iron wool: reacts very quickly without heating.
- HCl dissolved in water forms H^+ ions, so blue litmus turns red. HCl in methylbenzene does not ionise, so litmus stays blue.

MODEL ANSWER · HALOGEN DISPLACEMENT (6) · JUN 23
Bromine + potassium chloride: no colour change; the solution stays orange. Bromine cannot displace chlorine, so chlorine is more reactive than bromine.
Bromine + potassium iodide: the solution turns brown. Bromine displaces iodine ($Br_2 + 2KI \rightarrow 2KBr + I_2$), so bromine is more reactive than iodine.
Order: chlorine > bromine > iodine.

MODEL ANSWER · WHY DISPLACEMENT IS REDOX (2)
$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$: chlorine **gains electrons, so it is reduced**. Iodide ions **lose electrons, so they are oxidised**. Writing "iodine loses electrons" limits you to 1 mark.

Group 0

Group 0 elements are unreactive because they have a **full outer shell**, so they do not lose, gain or share electrons. Boiling point increases with A_r. Helium replaced hydrogen in airships: both have low density, but helium is unreactive while hydrogen is flammable/explosive in air.

7 Gases in the atmosphere and the % oxygen experiment

Dry air: nitrogen ≈ 78% ("about 80%" in multiple choice), oxygen ≈ 21%, argon ≈ 1%, carbon dioxide ≈ 0.04%. Air has no formula or M_r because it is a mixture.

Calculating the % oxygen

% O_2 = volume used up ÷ total starting volume × 100. The total must include the gas in the flask and tubes as well as the syringe.

Jun 22: $(90 - 22) \div (260 + 90) \times 100 = \mathbf{19.4\%}$ · Jan 21: $75 \div (265 + 100) \times 100 = \mathbf{20.5\%}$ · Nov 25 (water column): $(93 - 77) \div 93 \times 100 = \mathbf{17\%}$ · O_2/Ne mixture: $(75 - 30) \div 75 = \mathbf{60\%}$

- **Iron method:** the iron is wet because water is needed for rusting. The product is brown Fe_2O_3. Iron *powder* has a larger surface area, so it reacts faster.
- **Hot copper method:** copper is heated to give the particles enough energy to react. It turns black (copper(II) oxide). **All the oxygen has reacted** when the syringe reading stays constant.
- **Result too low:** the experiment was not left long enough, so leave it until the volume is constant. Other causes: a leak, or the gas not cooled to room temperature before reading.
- A 1 cm^3-division syringe cannot measure the CO_2 in air: 0.04% of 76 cm^3 is far below 1 cm^3.

Pollutant gases

Gas	How it forms	Problem (use this wording)
CO_2	complete combustion; thermal decomposition of carbonates ($CuCO_3 \rightarrow CuO + CO_2$, green → black)	greenhouse gas, causing global warming / climate change
CO	incomplete combustion (limited oxygen)	toxic: reduces the capacity of the blood to carry oxygen (binds to haemoglobin)
C (soot)	incomplete combustion	breathing / respiratory problems; black deposit on apparatus
SO_2	sulfur *impurity* in fuels burns: $S + O_2 \rightarrow SO_2$	dissolves in rainwater to form **acid rain**, which kills fish and plants and corrodes limestone buildings
NO_x	N_2 and O_2 *from the air* react at the high temperature in engines: $N_2 + O_2 \rightarrow 2NO$	acid rain: $4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$

> **TRAPS.** Petrol does *not* contain nitrogen; the nitrogen comes from the air (Jan 23). NO forms only at high temperature because the reaction has a high activation energy. Answers that link these gases to the ozone layer are penalised.

8 Chemical formulae, moles and every calculation type

Ions you must know

Positive	H^+, Li^+, Na^+, K^+, NH_4^+, Ag^+, Mg^{2+}, Ca^{2+}, Ba^{2+}, Zn^{2+}, Cu^{2+}, Fe^{2+}, Pb^{2+}, Fe^{3+}, Al^{3+}
Negative	OH^-, Cl^-, Br^-, I^-, NO_3^-, F^-, O^{2-}, S^{2-}, SO_4^{2-}, CO_3^{2-}, N^{3-}, PO_4^{3-}

Balance the charges. Examples set in past papers: $(NH_4)_3N$, $(NH_4)_2S$, Mg_3N_2, AlN, $Al_2(CO_3)_3$, $AlPO_4$, $Ca_3(PO_4)_2$, Li_2SO_4, $Mg(NO_3)_2$, $AlCl_3$, $ZnSO_4$, $(NH_4)_2CO_3$. Ammonium sulfate is $(NH_4)_2SO_4$ because NH_4^+ is 1+ and SO_4^{2-} is 2−, so two NH_4^+ ions balance one sulfate.

Counting: $Ca_3(PO_4)_2$ has 13 atoms. $Na_2SO_4 \cdot 7H_2O$ has 4 elements and 28 atoms. $Al_2(SO_4)_3$ has 3 elements and 17 atoms. $C_3H_5N_3O_9$ has 4 elements and 20 atoms.

The calculation toolkit

Type	Method	Past-paper example
% by mass	(A_r × number of atoms) ÷ M_r × 100	Ca in $Ca_3(PO_4)_2$: 120/310 = 38.7%; N in NH_4NO_3: 28/80 = 35%; N in NH_3 = 82.4%
Mass of element in a sample	sample mass × (element mass ÷ M_r)	N in 1.0 kg $(NH_4)_2SO_4$ = 1000 × 28/132 = 212 g
Empirical formula	% ÷ A_r (never ÷ atomic number); ÷ smallest; scale to whole numbers	Pb 90.7%, O 9.3% → 0.438 : 0.581 = 1 : 1.33 = 3 : 4 → Pb_3O_4
Molecular formula	M_r ÷ empirical formula mass = n	C_2H_4Cl (63.5), M_r 127 → $C_4H_8Cl_2$
Reacting masses	mass → moles → equation ratio → moles → mass	Fe_2O_3 + 3CO: 28 g Fe = 0.5 mol → 0.25 mol Fe_2O_3 → 40 g
Tonnes	work in megamoles (tonnes ÷ M_r); answer comes out in tonnes	892 t PbO ÷ 223 = 4 Mmol → 2 Mmol CO_2 × 44 = 88 t
Excess / limiting	use the equation to find the moles of one reactant needed for the other; compare with what is present	2Fe + 3Cl_2: 0.060 mol Cl_2 needs 0.040 mol Fe; 2.8 g = 0.050 mol, so Fe is in excess
Max product from mixture	find the limiting reagent first	2Mg + O_2: 0.50 mol Mg, 0.20 mol O_2 → O_2 limits → 0.40 mol MgO
% yield	actual ÷ theoretical × 100	21 g ÷ 28 g = 75%; over 100% means the product was still wet (not dried)
Moles of gas produced	moles fuel × total gas coefficient	3.7 g butanol = 0.05 mol × (2 CO_2 + 2 CO + 5 H_2O) = 0.45 mol
Combustion coefficients	mol CO_2 = C atoms; mol H_2O = H atoms ÷ 2; balance O_2 last	396 g CO_2 = 9; 180 g H_2O = 10 → C_9H_{20}; O_2: (18 + 10) ÷ 2 = 14
Find A_r of a metal	mass O lost → mol O = mol M (for MO) → A_r = mass M ÷ mol	4.46 − 4.14 = 0.32 g O = 0.020 mol; 4.14 ÷ 0.020 = 207 → lead

WORKED EXAMPLE · EXCESS AND REMAINING MASS (JAN 23, 3 MARKS)
Mg + 2HNO_3: 0.025 mol HNO_3 reacts with 0.0125 mol Mg = 0.30 g. Starting with 0.75 g, the Mg left over is 0.75 − 0.30 = **0.45 g**.

TRAPS. Dividing by atomic numbers, or inverting the division, scores 0 for empirical formula. "Show that" answers need full working, not just the final value. Write intermediate values to at least 3 s.f. and round only at the end.

9 Bonding, structure and properties

Ionic bonding

DEFINITIONS

Ionic bond: strong electrostatic attraction between oppositely charged ions.

Formation (e.g. $MgCl_2$): magnesium loses two electrons; each chlorine atom gains one electron. Configurations: Mg 2.8.2 → 2.8; Cl 2.8.7 → 2.8.8. Li_2O: each Li 2.1 → 2; O 2.6 → 2.8. Any mention of sharing electrons scores 0.

Diagrams of ions show the full shells, square brackets and the charge on each ion. Cl^- is 2.8.8; O^{2-} is 2.8.

MODEL ANSWER · HIGH MELTING POINT OF A GIANT IONIC COMPOUND (3)

Giant ionic lattice with **strong electrostatic forces of attraction between oppositely charged ions**, which need **a large amount of energy** to overcome.

MgO vs NaCl (5): both are giant ionic lattices with electrostatic attraction between oppositely charged ions. The ions in MgO have **higher charges** (Mg^{2+}, O^{2-} vs Na^+, Cl^-), so the attraction is stronger and more energy is needed to overcome it.

Conductivity: solid ionic compounds do not conduct because the ions are fixed in position and cannot move. They conduct when molten or in solution, because the ions can then move.

Covalent bonding

DEFINITION (2)

Electrostatic attraction between a **shared pair of electrons** and the **nuclei** of the two bonded atoms.

Molecule	Dot-and-cross essentials (outer shells)
H_2, Cl_2, F_2, HCl	1 shared pair; Cl and F each have 3 lone pairs
O_2; CO_2	O=O: 2 shared pairs, 2 lone pairs on each O. CO_2: two C=O double bonds
N_2	triple bond: 3 shared pairs, plus 1 lone pair on each N
H_2O; H_2O_2	H_2O: 2 bonding pairs, 2 lone pairs on O. H_2O_2: H–O–O–H, 3 bonding pairs, 2 lone pairs on each O
NH_3	3 N–H pairs and **1 lone pair** on N (this lone pair is the most often missed mark)
CH_4, C_2H_6, C_3H_8	only single pairs; every C has 4 shared pairs, with no lone pairs on C or H
C_2H_4; C_2H_3Cl	2 shared pairs between the carbons; Cl has 3 lone pairs
CH_3Cl; $CBrF_3$	4 bonding pairs around C; each halogen has 3 lone pairs

Structure types: identify from data (Jun 23)

Structure	Melting point	Conducts (solid)	Conducts (molten)	Examples
simple molecular (covalent)	low	no	no	CH_4, CO_2, SO_2, halogens, caffeine, C_{60}
giant covalent	very high	no (graphite: yes)	no	diamond, graphite, SiO_2
giant ionic	high	no	yes	NaCl, MgO, $CaBr_2$, PbS
metallic	usually high	yes	yes	Cu, Li

MODEL ANSWER · SIMPLE MOLECULAR: LOW BOILING POINT (2) · BIGGER MOLECULE BOILS HIGHER (3)

Weak **intermolecular forces** (forces between molecules) need **little energy** to overcome.
Hexane vs butane: hexane is a larger molecule (longer chain), so it has **stronger intermolecular forces** and more energy is needed to overcome them.

MODEL ANSWER · COMPARE A GIANT IONIC AND A SIMPLE MOLECULAR SUBSTANCE (5) · $CABR_2$ VS CAFFEINE, PBS VS SO_2

$CaBr_2$ has a **giant ionic lattice** with many strong electrostatic attractions between oppositely charged ions. Caffeine has a **simple molecular** structure with weak intermolecular forces between molecules. Much more energy is needed to break the electrostatic attractions than to overcome the intermolecular forces.

MODEL ANSWER · DIAMOND (3)

Giant covalent structure. Each carbon is bonded to four others in a rigid tetrahedral lattice. There are **many strong covalent bonds**, which need a large amount of energy to break. This gives a high melting point and makes diamond very hard, so it is used in cutting tools. It has no delocalised electrons, so it does not conduct.

MODEL ANSWER · GRAPHITE IS SOFT AND CONDUCTS (6) · M3 AND M7 ESSENTIAL

1. Carbon atoms are joined by strong covalent bonds, and each carbon is bonded to **three** others.
2. The atoms form **layers** (hexagonal sheets).
3. There are **weak forces between the layers**…
4. …so the layers can **slide over each other**, which makes graphite soft and suitable for pencils and lubricants.
5. Each carbon has one **delocalised electron**…
6. …and these electrons are **free to move** through the structure, so graphite conducts electricity.

Calling the forces between layers “intermolecular forces” loses that mark. Any mention of ions in graphite scores 0.

MODEL ANSWER · C_{60} FULLERENE MELTS FAR LOWER THAN DIAMOND/GRAPHITE (4)
C_{60} is simple molecular: only **weak intermolecular forces** need to be overcome, which takes little energy. Diamond and graphite are giant covalent: **many strong covalent bonds** must be broken, which takes much more energy.

10 Reactivity, redox and rusting

K > Na > Li > Ca > Mg > Al > (C) > Zn > Fe > (H) > Cu > Ag > Au

- **Ordering metals from a table:** a metal that reacts with water is the most reactive. Its reaction with acid is "not done" because it would be dangerous/explosive. A metal that reacts with neither water nor acid is Cu, Ag or Au. A slow reaction with acid only suggests Zn, Fe or Sn.
- Metal + acid → salt + hydrogen. Copper does not react with dilute acids because it is below hydrogen.
- Aluminium cannot be extracted with carbon because it is more reactive than carbon.

Displacement

Experiment	Observations
Mg (excess) in $CuSO_4(aq)$	brown/pink solid forms; blue solution turns colourless; temperature rises
Fe in $CuSO_4(aq)$	pink-brown coating of copper; solution turns from blue to pale green
Zn in $FeSO_4(aq)$	zinc turns dark grey (iron coating); green solution turns colourless
Cu in $MgSO_4(aq)$; Fe in $ZnSO_4(aq)$	no change: the metal is less reactive and cannot displace the other
Mg in $MgSO_4(aq)$	no change: a metal cannot displace itself

Temperature-rise method (Nov 25, Jan 20): the larger the temperature rise, the more reactive the metal. A rise of 0 °C means no reaction (e.g. silver, which is below copper). Control: the concentration and volume of the solution, the mass and surface area of the metal, the starting temperature and the rate of stirring. Copper(II) *carbonate* cannot replace copper(II) sulfate because it is insoluble.

Redox

DEFINITIONS
Oxidation = gain of oxygen or loss of electrons. **Reduction** = loss of oxygen or gain of electrons. Remember OIL RIG.
Oxidising agent = the species that gains electrons (and is itself reduced). **Reducing agent** = the species that loses electrons (and is itself oxidised).

- **Thermite:** $2Al + Fe_2O_3 \rightarrow 2Fe + Al_2O_3$. Aluminium is more reactive than iron, so it displaces it. Al gains oxygen (oxidised); Fe_2O_3 loses oxygen (reduced). The reaction is exothermic: heat energy is given out to the surroundings.
- **Zn + CuO → ZnO + Cu:** the oxidising agent is CuO, because it gives oxygen to zinc.
- **Zn + $FeSO_4$:** the oxidising agent is **Fe^{2+}**, which gains electrons from zinc. "Iron" or "Fe" is rejected.

- **3Mg + 2Al^{3+} → 3Mg^{2+} + 2Al:** the reducing agent is Mg, because it donates electrons to Al^{3+}.
- **Fe_2O_3 + 3CO and PbO + H_2:** the oxide loses oxygen and is reduced; CO or H_2 gains oxygen and is oxidised.

Reducing a metal oxide with hydrogen (Jun 21, Jan 24)

- Burn the excess hydrogen at the hole and relight it if it goes out. Hydrogen is flammable, and escaping gas could cause an explosion.
- Keep passing hydrogen until the tube is cool. This stops air getting in and re-oxidising the hot metal.
- Reheat, cool and reweigh **until the mass is constant**. This is the standard 2-mark answer.

Rusting

- Iron needs **oxygen (air) and water**. Rust is hydrated iron(III) oxide. Rusting is an **oxidation**.
- **Barrier methods** (paint, oil/grease, plastic coating, chromium plating) work only while the coating is intact.

MODEL ANSWERS · BARRIER (2) AND GALVANISING / SACRIFICIAL PROTECTION (2)
Paint: forms a barrier that stops oxygen and water reaching the iron.
Zinc (galvanising): zinc is **more reactive than iron**, so it oxidises/corrodes **instead of** the iron. This protection still works when the coating is scratched.

11 Acids, alkalis and salt preparation

Indicator	Acid	Alkali
litmus	red	blue
phenolphthalein	colourless	pink ("red/purple" rejected)
methyl orange	red	yellow (orange at neutral)
universal indicator	red (pH 0–3), orange/yellow	green at 7; blue/purple (pH 10–14)

- **Acid** = proton (H^+) donor. **Base** = proton acceptor. **Alkali** = a soluble base that releases OH^- ions in water.
- **Ammonia (Jun 26):** it is an alkali because it forms OH^- ions in water, and a base because it accepts H^+. Its pH is about 11, less alkaline than NaOH at about 14.
- Acid + metal → salt + H_2. Acid + base → salt + water. Acid + carbonate → salt + water + CO_2. $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$ is a neutralisation.
- A solution of $FeCl_3$ is acidic: litmus turns red, and the ion responsible is H^+.

Soluble	Insoluble
all Na, K and NH_4^+ salts; all nitrates; most chlorides; most sulfates	AgCl, $PbCl_2$; $BaSO_4$, $CaSO_4$, $PbSO_4$; carbonates and hydroxides (except Na, K, NH_4^+); CuO, PbO

Soluble salt from an insoluble solid (metal, oxide or carbonate) + acid

MODEL ANSWER · PURE, DRY CRYSTALS (5–6)

1. Warm the acid to increase the rate, then add the solid a spatula at a time with stirring.
2. Continue until the solid is in **excess**: solid remains and fizzing stops.
3. **Filter** to remove the excess solid.
4. **Heat the filtrate to evaporate some of the water** until it is saturated, i.e. crystals form on a glass rod dipped in and cooled.
5. **Leave to cool** so crystals form.
6. **Filter** off the crystals (or decant the liquid).
7. **Dry** them between filter papers, in a warm oven or in a desiccator. Never heat them directly or put them in a hot oven.

Heating to dryness scores only about M1. Leaving the solution with no heating loses the crystallisation-point mark. Washing after drying loses the drying mark.

- **Why excess solid?** So all the acid reacts and the product is not contaminated with acid. The excess is easy to remove by filtration because it is insoluble.
- **Why powder?** Larger surface area, so more frequent collisions and a faster reaction.
- Copper metal cannot be used to make $CuSO_4$ because it does not react with dilute acid.
- After filtering: excess solid is left on the filter paper; the salt solution (salt + water) is in the beaker.
- $CuSO_4$ solution is **blue**, and CuO is **black**.

Insoluble salts: precipitation

Mix two soluble salt solutions, filter, wash the residue with distilled water and dry. Examples: $Pb(NO_3)_2(aq) + 2KCl(aq) \rightarrow PbCl_2(s) + 2KNO_3(aq)$; $AgNO_3 + NaCl \rightarrow AgCl$(s, white). To obtain the *soluble* product (e.g. $NaNO_3$), filter off the precipitate and crystallise the filtrate. Using **equal moles** of reactants leaves no excess reagent to contaminate the product.

Precipitate-height graph (Jun 23): the first line passes through the origin because there is no precipitate with no reagent. The point where the two lines cross gives the volume needed for complete reaction. Anomalous point causes: precipitate not allowed to settle, height misread, or too much solution added.

12 Chemical tests

Test for	Method	Positive result
Li^+ / Na^+ / K^+ / Ca^{2+} / Cu^{2+}	flame test: clean wire dipped in conc. HCl, then in the solid, then held in a blue flame	red (crimson) / yellow / lilac / orange-red / blue-green
Cu^{2+}, Fe^{2+}, Fe^{3+}	add sodium hydroxide solution	blue ppt / green ppt / red-brown ppt
NH_4^+	add NaOH(aq) and warm; test the *gas* with damp red litmus	litmus turns blue (ammonia)
Cl^- / Br^- / I^-	add dilute **nitric acid**, then silver nitrate solution	white / cream / yellow precipitate
SO_4^{2-}	add dilute **HCl (or HNO_3)**, then barium chloride solution	white precipitate
CO_3^{2-}	add dilute acid; bubble the gas through limewater	effervescence; limewater turns milky
H_2 / O_2	lighted splint / glowing splint	squeaky pop / relights
CO_2	bubble through limewater	cloudy/milky ($CaCO_3$ forms)
Cl_2	damp blue litmus paper	turns red, then bleached white
NH_3	damp red litmus paper	turns blue
water	anhydrous copper(II) sulfate	white → blue

- **Why clean the wire?** To remove other ions that would mask or change the flame colour. The wire needs a high melting point and must be unreactive.
- **Why acidify before the sulfate test?** To remove carbonate ions, which would also give a white precipitate with barium ions.
- **4CH0 extension (Jan 19):** Ca^{2+}, Zn^{2+} and Al^{3+} all give white precipitates with NaOH. Only the Ca^{2+} precipitate remains in excess NaOH, so "white ppt = calcium" is not a valid conclusion.

MODEL ANSWER · IDENTIFY $FEBR_2$, $FECL_2$, $FEBR_3$ OR $FECL_3$ (6) · JAN 24

Cation: add sodium hydroxide solution. A green precipitate shows Fe^{2+} (iron(II)); a red-brown precipitate shows Fe^{3+} (iron(III)).

Anion: to a fresh sample, add dilute nitric acid then silver nitrate solution. A cream precipitate shows bromide; a white precipitate shows chloride.

The same template answers ammonium carbonate (Jun 24), ammonium sulfate (Jun 21), lithium carbonate (Jun 23) and copper/iron halides (Jan 20). For each ion: reagent → observation → conclusion.

MODEL ANSWER · TELL AMMONIUM CHLORIDE FROM AMMONIUM SULFATE (6) · NOV 25

Chloride: add dilute nitric acid, then silver nitrate. A white precipitate shows chloride.

Sulfate: to a fresh sample, add dilute hydrochloric acid, then barium chloride. A white precipitate shows sulfate.

Do not test for the ammonium ion: both compounds contain it, so the test cannot tell them apart.

MODEL ANSWER · FOUR POTASSIUM/SODIUM SALTS (6) · JAN 21

Flame test: only sodium chloride gives a yellow flame. Add acid: only potassium carbonate fizzes. To the other two, add dilute nitric acid then silver nitrate: KCl gives a white precipitate and KI gives a yellow precipitate.

13 Energetics

- **Exothermic:** heat energy is given out to the surroundings, the temperature rises and ΔH is negative. Examples: combustion, neutralisation, displacement, metal + acid, Li_2O + water, dissolving anhydrous $CuSO_4$.
- **Endothermic:** heat energy is taken in from the surroundings, the temperature falls and ΔH is positive. Examples: dissolving NH_4NO_3 (+24.2 kJ/mol), Na_2CO_3(s) + HCl, dissolving hydrated $CuSO_4$.

THE CALCULATION (4–5 MARKS)

1. **Q = m × c × ΔT**. Here *m* is the mass of **water or solution**, never the fuel. For mixed solutions, add the volumes (25 + 25 cm^3 = 50 g).
2. **n** = moles of the fuel burned (mass loss of the burner ÷ M_r), or of the limiting reagent.
3. **ΔH = −Q ÷ n**, then divide by 1000 to get kJ/mol. Add the sign: negative if the temperature rose.

Paper	Data	Working	Answer
Jun 23 pentanol	100 g water, ΔT 35.0 °C, 0.36 g, M_r 88	Q = 14 700 J; n = 0.0041 mol	−3600 kJ/mol
Jun 25 ethanol	100 g, 18 → 30 °C, 0.23 g, M_r 46	Q = 5040 J; n = 0.0050 mol	−1008 kJ/mol
Jun 24 Mg + HNO_3	40 g, ΔT 16.4 °C, 0.12 g Mg	Q = 2755 J; n = 0.005 mol	−550 kJ/mol (2 s.f.)
Jan 23 HCl + NaOH	Q = 2880 J, 0.0500 mol	2.880 kJ ÷ 0.0500	−57.6 kJ/mol
Nov 20 $CuSO_4$(s)	50 g, ΔT 3.3 °C, 1.70 g, M_r 159.5	Q = 693 J; n = 0.0107	−64.8 kJ/mol
Jun 19 NH_4NO_3	100 g, 20.0 → 14.2 °C, 8.00 g, c = 4.18	Q = 2420 J; n = 0.100	+24.2 kJ/mol

Reverse: find the maximum temperature (Nov 25, Jan 23). ΔT = Q ÷ (m × c) = 3780 ÷ (50 × 4.2) = 18 °C. Maximum temperature = 19 + 18 = **37 °C**. Remember to add the starting temperature (this step carries M4).

Errors and improvements (calorimetry)

Problem	Improvement / reason
heat lost to the surroundings, so the measured value is too small	polystyrene cup or lid (insulator, *reduces* heat loss); draught shield
incomplete combustion: black soot (carbon) because the oxygen supply is limited	(acknowledge as an error) the measured ΔH is less exothermic
glass beaker	use a copper/aluminium can, a better conductor of heat than glass
uneven temperature	stir so the temperature is the same throughout the water
fuel evaporates from the wick	weigh the burner **immediately** after extinguishing it
fair test between fuels	same volume of water, same distance from flame to can, same temperature rise, stirring
temperature read too early, or mixture not stirred (anomaly)	the highest temperature was not reached

Neutralisation graph with two crossing lines (Jun 25, Nov 25): the first line rises because the reaction releases heat. After neutralisation no further reaction happens, and the added cooler solution lowers the temperature. The **intersection** gives the volume needed for neutralisation and the maximum temperature.

Alcohols data (Jan 20): ΔH becomes more negative (more exothermic) by about 650 kJ/mol for each extra carbon atom, giving a straight-line trend. Extrapolate with a ruler.

14 Rates of reaction

MODEL ANSWERS · COLLISION THEORY

Concentration (3): more particles in the same volume, so **more frequent collisions** (more collisions per unit time), so the rate increases.
Surface area / powder (3): greater surface area, so more frequent collisions, so a faster rate.
Pressure (gases) (3): particles are closer together, so more collisions per unit time.
Temperature (3–4): particles gain kinetic energy and move faster. More collisions have energy ≥ the **activation energy**, so there are **more frequent successful collisions** and the rate increases. Temperature is the only factor where you mention energy.
Catalyst (2): provides an **alternative pathway** with a **lower activation energy**. It speeds up the reaction and is chemically unchanged at the end, so it is not used up.

MODEL ANSWER · EXPLAIN THE SHAPE OF A PRODUCT–TIME CURVE (4)

The curve is **steepest at the start** because the acid concentration is highest. It **becomes less steep** as the acid is used up and becomes more dilute, so collisions become less frequent. It **levels off** when the acid (the limiting reactant) is used up. Marble chips are left over because they were in excess. The rate decreases during the reaction because the H^+ concentration falls, giving fewer collisions per unit time.

Sketching a second curve

Change	Initial gradient	Final volume / mass
higher temperature, powder, catalyst, more concentrated *excess* reagent	steeper	same
lower temperature, lumps instead of powder	less steep	same
half the concentration (same volume) of the *limiting* reagent	less steep	**half**
half the mass of the limiting solid (e.g. Mg)	(similar or less steep)	**half**
double the limiting reagent + higher temperature	steeper	**double**

Always start at the origin, and level off at the exact value to within half a small square.

Apparatus and practical points

- **Gas volume:** gas syringe (or collection over water in an inverted measuring cylinder). Insert the bung quickly so little gas escapes, and start the stopwatch the moment the bung goes in.
- **Mass loss:** conical flask on a balance with a **cotton wool** plug. The plug lets CO_2 out but stops acid spray escaping, so the mass loss is due only to the gas. The balance reading falls because CO_2 escapes.
- Do not use H_2SO_4 with marble chips: insoluble $CaSO_4$ coats the chips and stops the reaction. Ca + H_2SO_4 stops for the same reason.
- Clean magnesium ribbon first, because the oxide layer slows the start of the reaction.
- A graduated beaker is acceptable for the acid when the acid is in excess, because a precise volume is not needed.
- **Thiosulfate:** the precipitate is sulfur. Time how long until the cross disappears; rate = 1 ÷ time. Above about 60 °C the solution cools during the run and the times become too short to measure accurately. Control the concentrations, the observer's viewing height and the same cross/flask.
- **Show MnO_2 is a catalyst:** filter it off after the reaction, dry it and weigh it. It still has the same mass (1.00 g).
- Variables to control: temperature, concentration and volume of solution, mass and surface area of the solid, mass of catalyst.

Rates from graphs

- **Rate at a given time:** draw a tangent touching the curve once at that time. Gradient = Δy ÷ Δx, with units cm^3/s or g/s. Without a tangent you can earn at most the units mark.
- **Mean rate:** total change ÷ time, e.g. 280 cm^3 ÷ 120 s = **2.33 cm^3/s**.
- **Read-off:** draw a vertical line from the time to the curve, then a horizontal line to the axis. Show both lines on the graph.
- Rate is **directly proportional** to concentration if the graph is a straight line through the origin (Jan 22). Rate vs temperature is a curve, so describe it as "positive correlation" and do not say "proportional".

15 Reversible reactions and water of crystallisation

- ⇌ means the reaction is **reversible**: it can go both forwards and backwards. Do not just say "equilibrium".
- $CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(l)$. Heating blue hydrated crystals in a boiling tube or evaporating basin turns them white, and colourless liquid (water) condenses. Adding water to the white anhydrous powder turns it blue.
- $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$.

FINDING X IN A HYDRATED SALT (4–5)

1. Mass of anhydrous salt = (tube + anhydrous) − empty tube.
2. Mass of water = (tube + hydrated) − (tube + anhydrous).
3. Find the moles of each (divide by M_r).
4. x = moles of water ÷ moles of anhydrous salt, rounded to a whole number.

Na_2SO_4 (Jun 23): 3.55 g → 0.025 mol; 4.50 g water → 0.25 mol; **x = 10**.
$MgSO_4$ (Jan 25): 6.3 g water → 0.35 mol; 6.0 g $MgSO_4$ → 0.05 mol; **x = 7**.
$KAl(SO_4)_2$ (Jun 24): 10.8 g → 0.6 mol; 12.9 g → 0.05 mol; **x = 12**.

- **Heat to constant mass:** reheat, cool and reweigh until two readings agree. This ensures all the water has been removed.
- **Ice around the collection tube:** cools the water vapour so it condenses and all the water is collected.
- **Calculated x too low** (4.12 instead of 5): not all the water was removed, because it was not heated for long enough. **Volume collected too low:** steam escaped, so cool the delivery tube or use a condenser.
- The decomposition is thermal decomposition: breaking down a substance by heating. Other examples: $2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$ (the CO_2 makes cakes rise), and $CuCO_3 \rightarrow CuO + CO_2$. A crucible lid stops solid spitting out, but stops the gases escaping easily.

16 Organic chemistry

Crude oil

MODEL ANSWER · FRACTIONAL DISTILLATION AND CRACKING (6) · JUN 26

1. Crude oil is **heated until it vaporises** and enters near the bottom of the fractionating column.
2. There is a **temperature gradient**: the column is hot at the bottom and cooler at the top.
3. Hydrocarbons condense at different heights according to their **boiling points**. Short-chain molecules rise higher; long chains condense near the bottom.
4. Crude oil contains **more long-chain hydrocarbons than are needed**, and there is a greater **demand** for short chains such as petrol.
5. **Cracking** uses a silica/alumina (zeolite) catalyst at **600–700 °C**.
6. It produces shorter-chain **alkanes** for fuels and **alkenes** for making polymers.

For a labelled diagram (Nov 25): show a tower with a furnace before the inlet, crude oil entering at the bottom, kerosene leaving partway up, a temperature label (hottest at the bottom) and at least three other fractions in the correct order.

Fraction (top → bottom)	Use
refinery gases	bottled gas for heating and cooking
gasoline (petrol)	fuel for cars
kerosene (paraffin)	fuel for aircraft
diesel	fuel for lorries, buses and cars
fuel oil	fuel for ships and power stations
bitumen	road surfacing and roofing

Moving down the column, chain length, boiling point and viscosity increase, the colour gets darker and the fraction becomes less flammable. Higher boiling point means longer molecules, stronger intermolecular forces and more energy needed to overcome them. **Balancing cracking equations:** conserve C and H, e.g. $C_{14}H_{30} \rightarrow C_8H_{18} + 3C_2H_4$; $C_{20}H_{42} \rightarrow 3C_4H_8 + C_8H_{18}$; $C_{15}H_{32} \rightarrow C_8H_{18} + C_2H_4 + C_5H_{10}$. A **fuel** is a substance that releases heat energy when burned.

Key definitions

Hydrocarbon: a compound containing hydrogen and carbon atoms *only*.
Saturated: contains only single carbon–carbon bonds, so no more atoms can be added and it cannot undergo addition reactions.
Unsaturated: contains a C=C double bond.
Homologous series: same general formula; same functional group; each member differs from the next by CH_2; similar chemical properties; *trend* in physical properties.
Functional group: the atom or group of atoms that determines a compound's chemical reactions.
Isomers: compounds with the same molecular formula but different structural (displayed) formulae.

- **Alkanes** C_nH_{2n+2}; **alkenes** C_nH_{2n}. Every alkene has empirical formula CH_2 because it has twice as many H atoms as C. Alkane empirical formulae differ: $C_2H_6 \rightarrow CH_3$, but $C_4H_{10} \rightarrow C_2H_5$.
- A molecule with two C=C bonds (isoprene C_5H_8, buta-1,3-diene C_4H_6) does not fit C_nH_{2n}.
- Names by number of carbons: 1 meth-, 2 eth-, 3 prop-, 4 but-, 5 pent-, 6 hex-, 8 oct-.

Isomers to be able to draw (show every bond)

Formula	Isomers
C_4H_{10}	butane $CH_3CH_2CH_2CH_3$; methylpropane $CH_3CH(CH_3)CH_3$
C_5H_{12}	pentane; methylbutane $CH_3CH(CH_3)CH_2CH_3$; dimethylpropane $C(CH_3)_4$
C_4H_8 (alkenes)	but-1-ene $CH_2{=}CHCH_2CH_3$; but-2-ene $CH_3CH{=}CHCH_3$ (cis/trans accepted); methylpropene $CH_2{=}C(CH_3)_2$. Cyclobutane is rejected when an alkene is asked for.
C_3H_6	propene; isomer with a different functional group = **cyclopropane**
C_2H_6	**no isomers**: there is only one possible arrangement of its atoms

> **TRAP.** Bending a straight chain on paper does not make a new isomer. Check the longest continuous carbon chain.

Reactions

- **Complete combustion** gives $CO_2 + H_2O$: $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$; $C_5H_{11}OH + 7½O_2 \rightarrow 5CO_2 + 6H_2O$.
- **Incomplete combustion** (limited oxygen) gives CO and/or C (soot) + H_2O: $C_8H_{18}(l) + 7O_2(g) \rightarrow 5CO(g) + 3C(s) + 9H_2O(l)$.

MODEL ANSWER · ETHANE VS ETHENE WITH BROMINE (5) · MAX 3 PER COMPOUND
Ethane: substitution reaction; needs **UV light**; bromine water stays orange (or decolourises very slowly); products are **bromoethane** + **hydrogen bromide**: $C_2H_6 + Br_2 \rightarrow C_2H_5Br + HBr$.
Ethene: addition reaction at room temperature; bromine water **decolourises immediately** (orange → colourless); product is **dibromoethane**: $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$. The C=C bond breaks.

- **Test for unsaturation:** add bromine water. It turns from orange to colourless with an alkene and stays orange with an alkane. Say "bromine water", not just "bromine", and do not mention UV.
- Substitution products: $CH_4 + Br_2 \rightarrow CH_3Br + HBr$; $C_3H_8 \rightarrow C_3H_7Br$; $C_5H_{12} \rightarrow C_5H_{11}Br$. Draw HBr as H–Br when a displayed formula is asked for. Multiple substitution is accepted if the equation is balanced.
- Bromine water + alkene can give a bromo-alcohol, e.g. C_3H_7OBr or C_2H_5OBr. These appear as empirical-formula calculations.

Addition polymers

- Monomer with C=C → long chain with only single bonds (saturated). **n** $CH_2{=}CH_2 \rightarrow -[CH_2{-}CH_2]_n-$. Draw the repeat unit with **extension bonds** through the brackets, and put n after the bracket.
- Poly(propene) repeat unit: $-CH_2-CH(CH_3)-$. Poly(chloroethene) (PVC): $-CH_2-CHCl-$. A C=C left inside the repeat unit scores 0.

- **Monomer vs polymer:** the polymer has a much longer chain, only single bonds, and is a solid where ethene is a gas.
- **Number of monomers** = M_r(polymer) ÷ M_r(monomer), e.g. 2 490 000 ÷ 62.5 ≈ 40 000.
- Cracking (alkenes) → polymers. Ethene → poly(ethene). Ethene + HCl/O_2 → chloroethene → PVC.

MODEL ANSWER · DISPOSAL PROBLEMS (4)
Landfill: polymers are inert and non-biodegradable, so they remain for a very long time and take up space (landfill sites are filling up).
Burning: releases greenhouse gases (CO_2), contributing to global warming, and toxic gases (CO; HCl from PVC).

17 Practical and graph skills

- **Plotting:** every point within ± half a small square. Use small neat crosses. Choose a scale that uses more than half the grid and goes up in sensible steps (e.g. starting at 6.0 and rising 2 per large square).
- **Lines:** draw a smooth curve of best fit or a ruled straight line. Never join the dots. Ignore the anomalous point and circle it if asked.
- **Two-line graphs** (neutralisation, precipitation): draw both lines with a ruler and make sure they **cross**. Read both coordinates of the intersection.
- **Reading values:** draw the construction lines on the graph. Read to the nearest small square.
- **Describing a relationship:** "as X increases, Y increases". Say "directly proportional" only for a straight line through the origin, and never say "linear" for a curve.
- **Anomaly causes in the mark schemes:** temperature read too early; not stirred; too much or too little reagent added; precipitate not settled; parallax error.
- **Apparatus names:** gas syringe; conical flask; measuring cylinder (or burette/pipette for liquids, never a gas syringe); evaporating basin; tripod; filter funnel; (Liebig) condenser; crucible on a pipeclay triangle; spirit burner; polystyrene cup.
- **Valid test:** change one variable and keep the others constant. Give *specific* variables ("volume of acid", "concentration of acid"), not just "amount".
- **Accuracy improvements:** insulate; use a lid; leave longer until readings are constant; use a more precise instrument; repeat and average (repeating on its own is often not credited as an accuracy improvement).
- **Safety:** eye protection and gloves for concentrated/corrosive solutions; fume cupboard for toxic gases; never test reactive metals such as K or Na with acid.
- **Legacy 4CH0 only (Jan 19):** titration. Swirl the flask; concordant titres are within 0.20 cm^3; mean of the concordant titres to 2 d.p.; moles = conc × vol ÷ 1000.

18 Final A* checklist

1. Underline the command word. **State/Give**: a short fact. **Describe**: what happens or what you do. **Explain**: what happens *because* of what, so link every point with "so" or "because". **Discuss**: both sides. **Show that**: full working to more significant figures than the target value.
2. Count the marks. A 3-mark "explain" needs three distinct, linked points.

3. For every structure question, decide **giant or simple** before writing a word.
4. For every rate question, use "**frequency of collisions**". Mention energy only for temperature.
5. For every ion test, write **reagent → observation → conclusion**, and start with a fresh sample for the second test.
6. For every calculation: **units, sign, sensible significant figures**, and an answer on the answer line (it takes priority over working elsewhere).
7. For every equation: correct formulae first, then balance. State symbols only when asked, but then all of them.
8. Give colours with their start and end points: *orange → colourless*, *white → blue*, *green → black*.
9. In multiple-choice items, cross out wrong options. The distractors repeat across papers: "iron oxide", "sacrificial protection" as the name of galvanising, and mixed-up NH_3/NH_4^+ formulae.
10. Leave 10 minutes to recheck calculations. Most lost marks at A* level come from arithmetic slips and a missing ΔH sign.